Fundamentals of Image Processing - Lab

H2.083

Equipment





Basler ace2 USB3



Camera

- Basler ace 2 R
- 5 MP, 60 FPS



- Get Images:
  - UI: Basler pylon viewer
  - Python: pypylon pip package

Stand

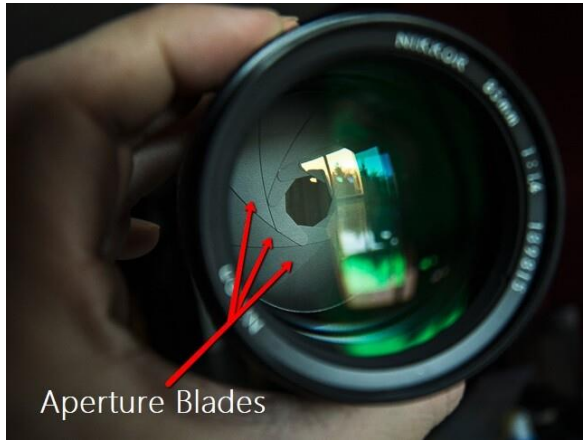


Adjust height



Lenses

5MP fixed focal length lenses (ranging from 5-75mm)



- Aperture Ring:
  - Controls aperture blades, which control how much light enters the camera.

- Focus Ring:
  - Controls the focusing distance of the image.
  - No change in focal length.



Lights

- Ring Lights
- Flat Lights





Materials for Tasks - I



- Coin Detection
  - Calibration Coin
  - bring your own coins

- SMD Detection
  - GPU



Materials for Tasks - II

- Event Detection
  - Pointing tools
  - Conyeor belt



- Gear Wheel Inspection
  - 3D printed gears

